

JEE MAIN REVISION MASTER: ELECTROCHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 226 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Nernst Equation & Cell Potential	84 PYQs	Core Formula
Very High	Electrode Potential & Delta G Relation	75 PYQs	Thermodynamics
Very High	Faraday's Laws of Electrolysis	53 PYQs	Quantitative Electrolysis
Very High	Electrolytic Conductance & Cell Constant	49 PYQs	Conductance
Very High	Commercial Batteries, Fuel Cells & Corrosion	48 PYQs	Applied Electrochemistry
Very High	Kohlrausch's Law & Weak Electrolyte Dissociation	41 PYQs	Electrolyte Behavior

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Nernst Equation & Cell Potential	$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log_{10} Q, \quad \Delta G^{\circ} = -n F E^{\circ}_{\text{cell}} = -2.303 RT \log_{10} K$
Electrode Potential & Delta G Relation	$\Delta G = -nFE, \quad \Delta G^{\circ}_3 = \Delta G^{\circ}_1 + \Delta G^{\circ}_2 \implies n_3 E^{\circ}_3 = n_1 E^{\circ}_1 + n_2 E^{\circ}_2$
Faraday's Laws of Electrolysis	$w = \frac{M}{F} \cdot I \cdot t, \quad \text{Moles of metal} = \frac{I \cdot t}{n F}, \quad F \approx 96500 \text{ C mol}^{-1}$
Electrolytic Conductance & Cell Constant	$\kappa = \frac{1}{R} \left(\frac{l}{A} \right) = G \cdot G^{\circ}, \quad \Lambda_m = \frac{1000 \cdot \kappa}{M} \quad (\kappa \text{ in S cm}^{-1})$
Commercial Batteries, Fuel Cells & Corrosion	Lead Storage Discharge: $\text{Pb(s)} + \text{PbO}_2\text{(s)} + 2\text{H}_2\text{SO}_4 \rightarrow 2\text{PbSO}_4\text{(s)} + 2\text{H}_2\text{O}$ $\eta = \frac{\Delta G}{\Delta H}$
Kohlrausch's Law & Weak Electrolyte Dissociation	$\Lambda_m^{\circ} = \nu_+ \lambda_+^{\circ} + \nu_- \lambda_-^{\circ}, \quad \alpha = \frac{\Lambda_m}{\Lambda_m^{\circ}}, \quad K_a = \frac{c \alpha^2}{1 - \alpha}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Nernst Equation & Cell Potential (Very High Yield — 84 PYQs)

Core Principle: Calculation of non-standard cell potential $E_{\text{cell}} = E^{\circ}_{\text{cell}} - (0.0591 / n) \log_{10} Q$ at 298 K, equilibrium constant K_{eq} , and concentration cells where $E^{\circ}_{\text{cell}} = 0$.

■ **Caution / JEE Pitfall:** 1. Forgetting to raise concentrations to their stoichiometric powers in reaction quotient Q . 2. Confusing oxidation vs reduction potential signs: $E^{\circ}_{\text{red}} = -E^{\circ}_{\text{ox}}$. 3. In concentration cells, anode is always the dilute compartment ($c_1 < c_2$).

■ **Speed Trick:** At 298 K, $E^{\circ} = (0.0591 / n) \log_{10} K$. For spontaneous cell operation, $E_{\text{cell}} > 0$ and $\Delta G < 0$.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "An electrochemical cell is constructed using half cells in the direction of spontaneous change: $\text{Fe}(\text{OH})_2(\text{s}) + 2\text{e}^- \rightarrow \text{Fe}(\text{s}) + 2\text{OH}^- (\text{aq})$..." **[Ans: N/A]**
- **[2026 Evening Shift] Q#1 (MCQ):** "For a closed circuit Daniell cell, which of the following plots is the accurate one at a given temperature?..." **[Ans: N/A]**
- **[2026 Evening Shift] Q#1 (Numerical):** "Consider the following electrochemical cell : $\text{Pt}|\text{O}_2(\text{g})(1\text{bar})|\text{HCl}(\text{aq})||\text{M}^{2+}(\text{aq}, 1.0\text{M})|\text{M}(\text{s})$ The pH above which, oxygen gas would start to..." **[Ans: 4]**

3.2 Electrode Potential & Delta G Relation (Very High Yield — 75 PYQs)

Core Principle: Relationship between standard reduction potentials and Gibbs free energy: $\Delta G^{\circ} = -nFE^{\circ}$. When combining half-reactions, always add extensive ΔG values, never add intensive E° potentials directly!

■ **Caution / JEE Pitfall:** Directly adding E° potentials of two half-cells to obtain E° of a composite couple (e.g. $\text{Fe}^{3+} \rightarrow \text{Fe}$ from $\text{Fe}^{3+} \rightarrow \text{Fe}^{2+}$ and $\text{Fe}^{2+} \rightarrow \text{Fe}$). You MUST weight by electron transfer count: $n_3 E_3^{\circ} = n_1 E_1^{\circ} + n_2 E_2^{\circ}$!

■ **Speed Trick:** Always write $\Delta G = -nFE$ before combining half-reactions. A more positive standard reduction potential E° denotes a stronger oxidizing agent (more readily reduced).

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Given at 298 K : $E_{\text{cell}}(\text{Fe}^{2+}/\text{Fe}) = X\text{Volt}$ $E_{\text{cell}}(\text{Fe}^{3+}/\text{Fe}) = Y\text{Volt}$ The $E_{\text{cell}}(\text{Fe}^{3+}/\text{Fe}^{2+})$ is..." **[Ans: B]**
- **[2026 Morning Shift] Q#1 (MCQ):** "An electrochemical cell is constructed using half cells in the direction of spontaneous change: $\text{Fe}(\text{OH})_2(\text{s}) + 2\text{e}^- \rightarrow \text{Fe}(\text{s}) + 2\text{OH}^- (\text{aq})$..." **[Ans: N/A]**
- **[2026 Evening Shift] Q#1 (Numerical):** "Consider the following electrochemical cell : $\text{Pt}|\text{O}_2(\text{g})(1\text{bar})|\text{HCl}(\text{aq})||\text{M}^{2+}(\text{aq}, 1.0\text{M})|\text{M}(\text{s})$ The pH above which, oxygen gas would start to..." **[Ans: 4]**

3.3 Faraday's Laws of Electrolysis (Very High Yield — 53 PYQs)

Core Principle: First and second laws of quantitative electrolysis: Mass deposited $w = Z I t = (M \cdot I \cdot t / n F)$. Total charge passed $Q = I t = n_e F$.

■ **Caution / JEE Pitfall:** 1. Using an incorrect n-factor (electrons transferred per molecule). E.g. for O_2 from H_2O , $n = 4$; for H_2 , $n = 2$. 2. Time t must strictly be substituted in seconds (s).

■ **Speed Trick:** Gas volume at STP: Moles $\times 22.4$ L. For electroplating, Volume = Area $\times$ Thickness and Mass = Density $\times$ Volume.

Representative Source PYQs:

- **[2026 Evening Shift] Q#3 (Numerical):** "Consider the following two half-cell reactions along with the standard reduction potential given : $CO_2 + 6H^+ + 6e^- \rightarrow CH_4$..." **[Ans: 560]**
- **[2026 Morning Shift] Q#4 (Numerical):** "Electricity is passed through an acidic solution of Cu^{2+} till all the Cu^{2+} was exhausted, leading to the deposition of 300 mg of C..." **[Ans: 111]**
- **[2025 Morning Shift] Q#1 (MCQ):** "A solution of aluminium chloride is electrolysed for 30 minutes using a current of 2 A . The amount of the aluminium deposited at the cathod..." **[Ans: N/A]**

3.4 Electrolytic Conductance & Cell Constant (Very High Yield — 49 PYQs)

Core Principle: Specific conductivity $\kappa = (1 / R) \left(l / A \right) = G \cdot G^*$. Molar conductivity $\Lambda_m = (1000\kappa / M)$ (with κ in $S\ cm^{-1}$) or $\Lambda_m = (\kappa / 1000M)$ (with κ in $S\ m^{-1}$).

■ **Caution / JEE Pitfall:** Unit confusion between cm and m. If κ is in $S\ m^{-1}$, use $\Lambda_m = (\kappa / c)$ (c in $mol\ m^{-3}$). If in $S\ cm^{-1}$, use $\Lambda_m = (1000\kappa / M)$ (M in $mol\ L^{-1}$).

■ **Speed Trick:** Cell constant $G^* = (l / A)$ is an intrinsic geometric property of the conductivity cell and is strictly independent of the electrolyte filled.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (Numerical):** "At 298 K , the molar conductivity of $x\%(w/w)MX$ solution (aqueous) is $123.5\ S\ cm^2\ mol^{-1}$. The conductance of same solution is $1.9\ \Omega^{-1}$..." **[Ans: 15]**
- **[2026 Evening Shift] Q#1 (Numerical):** "For strong electrolyte Λ_m increases slowly with dilution and can be represented by the equation $\Lambda_m = \Lambda_m^\circ - \dots$ " **[Ans: 4]**
- **[2026 Morning Shift] Q#3 (MCQ):** "Consider the following data. $\begin{array}{l} \text{Electrolyte} \\ \wedge_m^\circ \end{array} \begin{array}{l} (S\ cm^2\ mol^{-1}) \\ \end{array} \parallel BaCl_2 \ \& \ x_1 \parallel H_2S \dots$ " **[Ans: A]**

3.5 Commercial Batteries, Fuel Cells & Corrosion (Very High Yield — 48 PYQs)

Core Principle: Primary cells (dry cell, mercury cell), secondary lead-acid batteries (charge/discharge cycles), H_2-O_2 fuel cells (thermodynamic efficiency $\eta = (\Delta G / \Delta H)$), and electrochemical corrosion prevention.

■ **Caution / JEE Pitfall:** In a lead storage battery, the density of H_2SO_4 decreases during discharge because sulfuric acid is consumed and water is produced.

■ **Speed Trick:** In galvanic cells: Anode is negative, Cathode is positive (An Ox, Red Cat). In electrolytic cells: Anode is positive, Cathode is negative.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "An electrochemical cell is constructed using half cells in the direction of spontaneous change: $Fe(OH)_2(s) + 2e^- \rightarrow Fe(s) + 2OH^-(aq)$... " **[Ans: N/A]**
- **[2026 Evening Shift] Q#1 (Numerical):** "Consider the following electrochemical cell : $Pt|O_2(g)(1bar)|HCl(aq)||M^{2+}(aq,1.0M)|M(s)$ The pH above which, oxygen gas would start to..." **[Ans: 4]**
- **[2026 Evening Shift] Q#2 (MCQ):** "For a general redox reaction $Anode\ Red_1 \rightarrow Ox_1 + n_1e^-$ Cathode $Ox_2 + n_2e^- \rightarrow Red_2$..." **[Ans: N/A]**

3.6 Kohlrausch's Law & Weak Electrolyte Dissociation (Very High Yield — 41 PYQs)

Core Principle: Independent migration of ions at infinite dilution: $\Lambda_m^\circ = \nu_+ \lambda_{+}^\circ + \nu_- \lambda_{-}^\circ$. Degree of dissociation $\alpha = (\Lambda_m / \Lambda_m^\circ)$, and Ostwald's dilution law $K_a = (c\alpha^2 / 1 - \alpha)$.

■ **Caution / JEE Pitfall:** 1. Forgetting to multiply ionic conductances by stoichiometric coefficients (e.g., for $Al_2(SO_4)_3$ implies $2\lambda^\circ(Al^{3+}) + 3\lambda^\circ(SO_4^{2-})$). 2. For weak electrolytes, if $\alpha \ll 1$, do not approximate $1 - \alpha \approx 1$.

■ **Speed Trick:** Λ_m° for weak electrolytes like CH_3COOH can be obtained using strong electrolytes: $\Lambda_m^\circ(CH_3COOH) = \Lambda_m^\circ(CH_3COONa) + \Lambda_m^\circ(HCl) - \Lambda_m^\circ(NaCl)$.

Representative Source PYQs:

- **[2026 Morning Shift] Q#3 (MCQ):** "Consider the following data. $\text{Electrolyte} \wedge_m^\circ$ & χ_{12} ..." **[Ans: A]**
- **[2026 Morning Shift] Q#3 (MCQ):** "In the given electrochemical cell, $Ag(s)|AgCl(s)|FeCl_2(aq), FeCl_3(aq)|Pt(s)$ at 298 K, the cell potential ($E_{(cell)}$) will ..." **[Ans: N/A]**
- **[2026 Evening Shift] Q#3 (Numerical):** "Molar conductivity of a weak acid HQ of concentration 0.18 M was found to be 1/30 of the molar conductivity of another weak acid HZ with $c...$ " **[Ans: 2]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Given at 298 K : $E_{(Fe^{2+}/Fe)}^\circ = X \text{ Volt}$ $E_{(Fe^{3+}/Fe)}^\circ = Y \text{ Volt}$ The $E_{(Fe^{3+}/Fe^{2+})}^\circ$ in Volt at 298 K is given by :

- (A) $2X-3Y$
- (B) $3Y-2X$ **[Correct]**
- (C) $3Y+2X$
- (D) $Y+X$

Correct Answer: B

Solution Analysis: (i) $Fe^{2+}(aq) + 2e^- \rightarrow Fe(s)$ ΔG_1° (ii) $Fe^{3+}(aq) + 3e^- \rightarrow Fe(s)$ ΔG_2° (iii) $Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$ ΔG_3° iii=ii-i $-1 \times F \times E_{(Fe^{3+}/Fe^{2+})}^\circ = -3 \times F \times Y - (-2 \times F \times X)$ $E_{(Fe^{3+}/Fe^{2+})}^\circ = 3Y - 2X$ Option (B) is correct

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Evening Shift (Q#1)

At 298 K, the molar conductivity of x%(w/w)MX solution (aqueous) is $123.5 \text{ Scm}^2 \text{ mol}^{-1}$. The conductance of same solution is $1.9 \times 10^{-3} \text{ S}$. The value of x is $\times 10^{-2}$. (Given: cell constant $= 1.3 \text{ cm}^{-1}$; molar mass of MX is 75 g mol^{-1} , density of aqueous solution of MX at 298 K is 1.0 g mL^{-1})

Correct Answer: 15

Solution Analysis: Conductivity (κ) is related to the conductance (G) and the cell constant (G^*) by the following relationship: $\kappa = G \times G^*$. Substituting the given values: $\kappa = (1.9 \times 10^{-3} \text{ S}) \times (1.3 \text{ cm}^{-1})$ $\kappa = 2.47 \times 10^{-3} \text{ Scm}^{-1}$. The molar conductivity (Λ_m) is related to the conductivity (κ) and the molarity (C in mol L^{-1}) by the formula: $\Lambda_m = (\kappa \times 1000 / C)$. Rearranging the formula to solve for molarity (C): $C = \frac{\kappa \times 1000}{\Lambda_m}$. Substituting the values into the equation: $C =$

Benchmark Problem 3: JEE Main 2026 (Online) 22nd January Evening Shift (Q#1)

Consider the following electrochemical cell: $\text{Pt} | \text{O}_2(\text{g}) (1 \text{ bar}) | \text{HCl}(\text{aq}) || \text{M}^{2+}(\text{aq}, 1.0 \text{ M}) | \text{M}(\text{s})$. The pH above which, oxygen gas would start to evolve at anode is ____ (nearest integer). [Given : $E_{(\text{M}^{2+}/\text{M})}^\circ = 0.994 \text{ V}$, $E_{(\text{O}_2/\text{H}_2\text{O})}^\circ = 1.23 \text{ V}$ standard reduction potential and $(RT/F)(2.303) = 0.059 \text{ V}$ at the given condition]

Correct Answer: 4

Solution Analysis: For the oxygen electrode (reduction reaction): $\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}(\text{l})$. Nernst equation: $E_{(\text{O}_2/\text{H}_2\text{O})} = E_{(\text{O}_2/\text{H}_2\text{O})}^\circ - (0.059/4) \log(\frac{1}{p_{(\text{O}_2)}[\text{H}^+]^4})$. Given $p_{(\text{O}_2)} = 1 \text{ bar}$, so $\log p_{(\text{O}_2)} = 0$: $E_{(\text{O}_2/\text{H}_2\text{O})} = 1.23 + 0.059 \log[\text{H}^+]$. Since $\log[\text{H}^+] = -\text{pH}$: $E_{(\text{O}_2/\text{H}_2\text{O})} = 1.23 - 0.059 \text{ pH}$. For the metal electrode: $\text{M}^{2+} + 2\text{e}^- \rightarrow \text{M}(\text{s})$. Given $[\text{M}^{2+}] = 1.0 \text{ M}$, hence $E_{(\text{M}^{2+}/\text{M})} = E_{(\text{M}^{2+}/\text{M})}^\circ = 0.994 \text{ V}$. Oxygen will start evolving at the anode when the oxygen electrode becomes the anode, i.e. when its reduction pote

Benchmark Problem 4: JEE Main 2026 (Online) 24th January Morning Shift (Q#4)

Electricity is passed through an acidic solution of Cu^{2+} till all the Cu^{2+} was exhausted, leading to the deposition of 300 mg of Cu metal. However, a current of 600 mA was continued to pass through the same solution for another 28 minutes by keeping the total volume of the solution fixed at 200 mL. The total volume of oxygen evolved at STP during the entire process is ____ mL. (Nearest integer) [Given: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$, $E_{(\text{red})}^\circ = +0.34 \text{ V}$, $\text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$, $E_{(\text{red})}^\circ = +1.23 \text{ V}$. Molar mass of Cu = 63.54 g mol^{-1} . Molar mass of O_2 = 32 g mol^{-1} . Faraday Constant = 96500 C mol^{-1} . Molar volume at STP = 22.4 L]

Correct Answer: 111

Solution Analysis: Eq of Cu = Eq of O_2 . In electrolysis, the number of equivalents deposited and liberated are equal for the same quantity of electricity. For copper deposition: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$, so n-factor = 2. For oxygen evolution (reverse of given reduction): $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$, so n-factor = 4. So, equivalents of Cu deposited = equivalents of O_2 evolved during the first part. $\frac{300 \times 10^{-3} \times 263.54}{n_{(\text{O}_2)} \times 4} = 2.36 \times 10^{-3} = n_{(\text{O}_2)}$. When current is further passed After all Cu^{2+} is exhausted, the current no